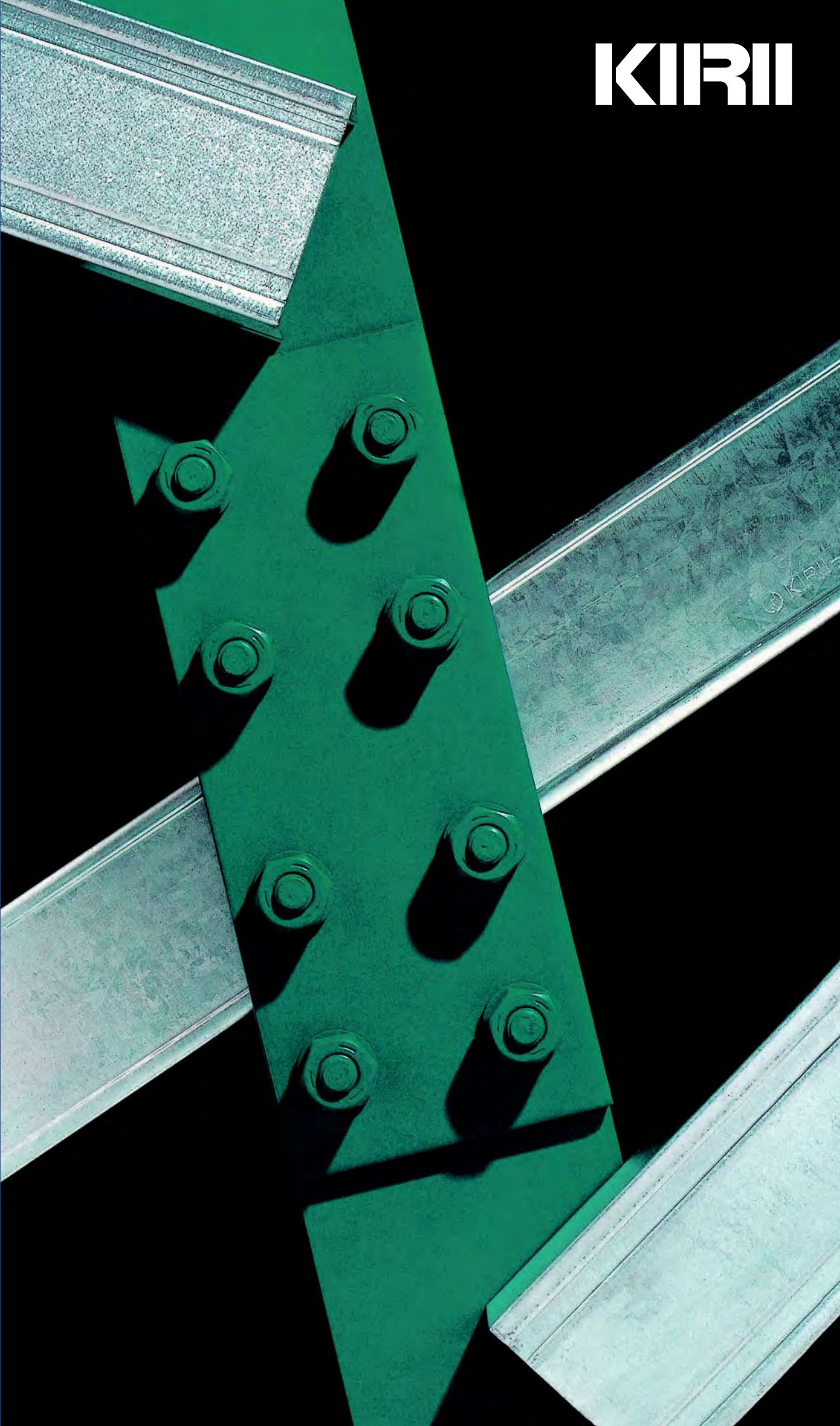


**KIRII**

# STUD60

Partition Steel Framing System







滋賀工場  
Shiga Factory



札幌工場  
Sapporo Factory



茨城工場  
Ibaragi Factory



九州工場  
kyushu Factory



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## Partition Steel Framing System

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## Introduction

**STUDCO** Partition Steel Framing System is used in a variety of interior, non-loadbearing construction applications. It consists of studs and runners are C-shaped members, roll-formed from corrosion resistant, galvanized steel. These members are assembled to provide efficient, noncombustible, low cost framing for partition walls, ceilings and column fireproofing.

Members comply with ASTM C645 & C955 and are manufactured by Kirii's accredited ISO 9001 factory.

When it comes to fire-rated system, CIKV. Framing System is UL-approved for the UL design assemblies including drywall and chase wall.

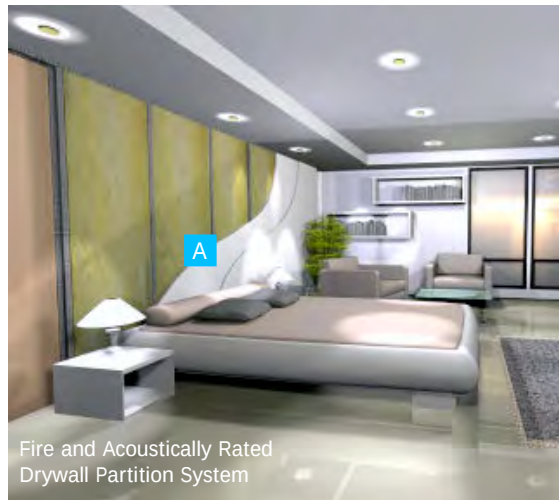


## Steel Framing Members

**STUDCO** Steel Studs are available in 1 $\frac{3}{4}$ ", 2", 2 $\frac{1}{2}$ ", 3", 3 $\frac{5}{8}$ ", 4", 5" and 6" web widths to accommodate in wide range of applications (see table 1). All studs have flanges that are knurled for quick screw attachment of gypsum wallboards.

Studs may be spaced at 12", 16" or 24" on-center based on wall height. Wall height, stud spacing, framing condition, deflection limitations and design load level are all factors that must be considered in determining the correct size and thickness of the applicable product.

**STUDCO** Steel Runner is available in corresponding sizes and is offered in a variety of flange heights including 1", 1 $\frac{1}{4}$ ", 1 $\frac{3}{4}$ " and 2" flanges (see table 2). Runner is used to align and secure studs to floors and ceilings, and is also used in conjunction with studs for constructing openings, headers and sills. Steel studs are twisted into top and bottom runner, generally are friction fit (no fasteners) except at doors, windows, etc.



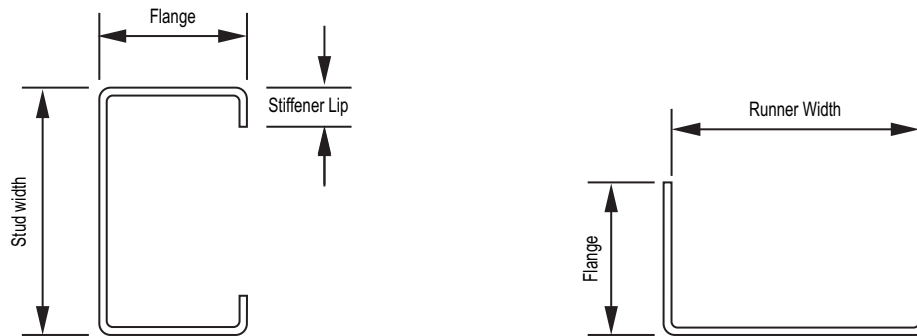


Table 1 : Availability of stud dimensions and steel thicknesses

| Product Code | Stud Width |      | Flange Height |    |           |    |           |    |           |    | Uncoated Steel Thickness (mm) |     |     |     |     |     |     |
|--------------|------------|------|---------------|----|-----------|----|-----------|----|-----------|----|-------------------------------|-----|-----|-----|-----|-----|-----|
|              |            |      | inch (mm)     |    | inch (mm) |    | inch (mm) |    | inch (mm) |    |                               |     |     |     |     |     |     |
|              | inch       | (mm) | 1 1/4         | 32 | 1 3/4     | 45 | 2         | 50 | 3         | 76 | 0.5                           | 0.6 | 0.8 | 1.0 | 1.2 | 1.5 | 2.0 |
| WS-134       | 1 3/4      | 45   | ●             |    |           |    |           |    |           |    | ●                             |     |     |     |     |     |     |
| WS-200       | 2          | 50   | ●             |    | ●         |    |           |    |           |    | ●                             | ●   | ●   | ●   | ●   | ●   |     |
| WS-212       | 2 1/2      | 64   | ●             |    | ●         |    | ●         |    |           |    | ●                             | ●   | ●   | ●   | ●   | ●   |     |
| WS-300       | 3          | 76   | ●             |    | ●         |    | ●         |    |           |    | ●                             | ●   | ●   | ●   | ●   | ●   |     |
| WS-358       | 3 5/8      | 92   | ●             |    | ●         |    | ●         |    |           |    | ●                             | ●   | ●   | ●   | ●   | ●   |     |
| WS-400       | 4          | 102  | ●             |    | ●         |    | ●         |    | ●         |    | ●                             | ●   | ●   | ●   | ●   | ●   |     |
| WS-500       | 5          | 127  |               |    | ●         |    | ●         |    | ●         |    |                               |     | ●   | ●   | ●   | ●   |     |
| WS-600       | 6          | 152  |               |    | ●         |    | ●         |    | ●         |    |                               |     | ●   | ●   | ●   | ●   |     |
| WS-800       | 8          | 200  |               |    |           |    | ●         |    |           |    |                               |     |     |     |     | ●   |     |
| * WS-400     | 4          | 102  |               |    |           |    |           |    | ●         |    |                               |     |     |     |     |     | ●   |
| * WS-500     | 5          | 127  |               |    |           |    |           |    | ●         |    |                               |     |     |     |     |     | ●   |
| * WS-600     | 6          | 152  |               |    |           |    |           |    | ●         |    |                               |     |     |     |     |     | ●   |

Non-standard width of stud can be customized to suit individual design and application with an agreed quantity

\* Subject to minimum order and agreed quantity.

Table 2 : Availability of runner dimensions and steel thicknesses

| Product Code | Runner Width |      | Flange Height |    |           |    |           |    |           |    | Uncoated Steel Thickness (mm) |     |     |     |     |     |     |
|--------------|--------------|------|---------------|----|-----------|----|-----------|----|-----------|----|-------------------------------|-----|-----|-----|-----|-----|-----|
|              |              |      | inch (mm)     |    | inch (mm) |    | inch (mm) |    | inch (mm) |    |                               |     |     |     |     |     |     |
|              | inch         | (mm) | 1             | 25 | 1 1/4     | 32 | 1 3/4     | 45 | 2         | 50 | 0.5                           | 0.6 | 0.8 | 1.0 | 1.2 | 1.5 | 2.0 |
| WR-134       | 1 3/4        | 45   | ●             |    | ●         |    |           |    |           |    | ●                             |     |     |     |     |     |     |
| WR-200       | 2            | 50   | ●             |    | ●         |    | ●         |    | ●         |    | ●                             | ●   | ●   | ●   | ●   | ●   |     |
| WR-212       | 2 1/2        | 64   | ●             |    | ●         |    | ●         |    | ●         |    | ●                             | ●   | ●   | ●   | ●   | ●   |     |
| WR-300       | 3            | 76   | ●             |    | ●         |    | ●         |    | ●         |    | ●                             | ●   | ●   | ●   | ●   | ●   |     |
| WR-358       | 3 5/8        | 92   | ●             |    | ●         |    | ●         |    | ●         |    | ●                             | ●   | ●   | ●   | ●   | ●   |     |
| WR-400       | 4            | 102  | ●             |    | ●         |    | ●         |    | ●         |    | ●                             | ●   | ●   | ●   | ●   | ●   | ●   |
| WR-500       | 5            | 127  |               |    | ●         |    | ●         |    | ●         |    |                               |     | ●   | ●   | ●   | ●   | ●   |
| WR-600       | 6            | 152  |               |    | ●         |    | ●         |    | ●         |    |                               |     | ●   | ●   | ●   | ●   | ●   |
| WR-800       | 8            | 200  |               |    |           |    |           |    | ●         |    |                               |     |     |     |     | ●   |     |

Deep flange runners are also available in 76mm and 100mm high for deflection head constructions.

## Weight of zinc coating

| Uncoated Steel Thickness (mm) | Coating Designation |        |            |        |           |          |
|-------------------------------|---------------------|--------|------------|--------|-----------|----------|
|                               | JIS G3302           |        | ASTM A653M |        | ASTM A653 |          |
|                               | Symbol              | (g/m2) | Symbol     | (g/m2) | Symbol    | (oz/ft2) |
| 0.5 - 1.5                     | Z 18                | 180    | Z 180      | 180    | G 60      | 0.60     |
| * 0.8 - 2.0                   | Z 27                | 275    | Z 275      | 275    | G 90      | 0.90     |

The coating masses shown are the industry standard for roll-formed steel from 0.5 to 1.5mm thick.

\* Z275/G90 available as special order.



## Limiting Heights:

1. Limiting heights computed are based on the theoretical properties of steel stud and deflection limits of L/120; L/240; and L/360 resulted from a uniform load of 5psf (240Pa) on partition surface, per ASTM C754 and GA-600 Gypsum Association.
2. Limiting heights are in accordance with AISI Standard "North American Specification For The Design of Cold-formed Steel" using all steel non-composite design.
3. Limiting heights are established by considering flexure, shear, web crippling and deflection, based on continuous support of each flange over the full length of the stud; with unperforated section properties; and full lateral support to both flanges in flexure.
4. Applications involving multiple spans, cantilevers, concentrated loads, etc should be investigated separately.



| 0.5mm member |      |               |      |                     | Limiting heights (mm)      |       |       |
|--------------|------|---------------|------|---------------------|----------------------------|-------|-------|
| Stud Width   |      | Flange Height |      | Uncoated Steel Thk. | 5 psf (240Pa) Uniform Load |       |       |
| inch         | (mm) | inch          | (mm) | mm                  | Deflection Limit           |       |       |
|              |      |               |      |                     | L/120                      | L/240 | L/360 |
| 2            | 50   | 1 1/4         | 32   | 0.5                 | 12                         | 305   | 3230  |
| 2            | 50   | 1 1/4         | 32   | 0.5                 | 16                         | 406   | 2800  |
| 2            | 50   | 1 1/4         | 32   | 0.5                 | 24                         | 610   | N/A   |
| 2 1/2        | 64   | 1 1/4         | 32   | 0.5                 | 12                         | 305   | 3680  |
| 2 1/2        | 64   | 1 1/4         | 32   | 0.5                 | 16                         | 406   | 3190  |
| 2 1/2        | 64   | 1 1/4         | 32   | 0.5                 | 24                         | 610   | 2600  |
| 3            | 76   | 1 1/4         | 32   | 0.5                 | 12                         | 305   | 4060  |
| 3            | 76   | 1 1/4         | 32   | 0.5                 | 16                         | 406   | 3520  |
| 3            | 76   | 1 1/4         | 32   | 0.5                 | 24                         | 610   | 2870  |
| 3 5/8        | 92   | 1 1/4         | 32   | 0.5                 | 12                         | 305   | 4490  |
| 3 5/8        | 92   | 1 1/4         | 32   | 0.5                 | 16                         | 406   | 3890  |
| 3 5/8        | 92   | 1 1/4         | 32   | 0.5                 | 24                         | 610   | 2980  |
| 4            | 102  | 1 1/4         | 32   | 0.5                 | 12                         | 305   | 4720  |
| 4            | 102  | 1 1/4         | 32   | 0.5                 | 16                         | 406   | 4090  |
| 4            | 102  | 1 1/4         | 32   | 0.5                 | 24                         | 610   | 2930  |



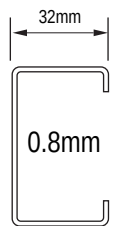
| 0.6mm member |      |               |      |                     | Limiting heights (mm)      |       |       |
|--------------|------|---------------|------|---------------------|----------------------------|-------|-------|
| Stud Width   |      | Flange Height |      | Uncoated Steel Thk. | 5 psf (240Pa) Uniform Load |       |       |
| inch         | (mm) | inch          | (mm) | mm                  | Deflection Limit           |       |       |
|              |      |               |      |                     | L/120                      | L/240 | L/360 |
| 2            | 50   | 1 1/4         | 32   | 0.6                 | 12                         | 305   | 3650  |
| 2            | 50   | 1 1/4         | 32   | 0.6                 | 16                         | 406   | 3160  |
| 2            | 50   | 1 1/4         | 32   | 0.6                 | 24                         | 610   | 2580  |
| 2 1/2        | 64   | 1 1/4         | 32   | 0.6                 | 12                         | 305   | 4160  |
| 2 1/2        | 64   | 1 1/4         | 32   | 0.6                 | 16                         | 406   | 3610  |
| 2 1/2        | 64   | 1 1/4         | 32   | 0.6                 | 24                         | 610   | 2940  |
| 3            | 76   | 1 1/4         | 32   | 0.6                 | 12                         | 305   | 4600  |
| 3            | 76   | 1 1/4         | 32   | 0.6                 | 16                         | 406   | 3980  |
| 3            | 76   | 1 1/4         | 32   | 0.6                 | 24                         | 610   | 3250  |
| 3 5/8        | 92   | 1 1/4         | 32   | 0.6                 | 12                         | 305   | 5090  |
| 3 5/8        | 92   | 1 1/4         | 32   | 0.6                 | 16                         | 406   | 4410  |
| 3 5/8        | 92   | 1 1/4         | 32   | 0.6                 | 24                         | 610   | 3600  |
| 4            | 102  | 1 1/4         | 32   | 0.6                 | 12                         | 305   | 5370  |
| 4            | 102  | 1 1/4         | 32   | 0.6                 | 16                         | 406   | 4650  |
| 4            | 102  | 1 1/4         | 32   | 0.6                 | 24                         | 610   | 3790  |



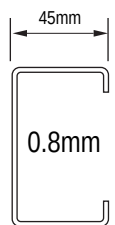
## Limiting Heights:

1. Limiting heights computed are based on the theoretical properties of steel stud and deflection limits of L/120; L/240; and L/360 resulted from a uniform load of 5psf (240Pa) on partition surface, per ASTM C754 and GA-600 Gypsum Association.
2. Limiting heights are in accordance with AISI Standard "North American Specification For The Design of Cold-formed Steel" using all steel non-composite design.
3. Limiting heights are established by considering flexure, shear, web crippling and deflection, based on continuous support of each flange over the full length of the stud; with unperforated section properties; and full lateral support to both flanges in flexure.
4. Applications involving multiple spans, cantilevers, concentrated loads, etc should be investigated separately.

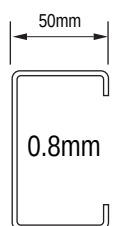
| 0.8mm member |      |               |      |                     |              | Limiting heights (mm)      |                  |       |       |
|--------------|------|---------------|------|---------------------|--------------|----------------------------|------------------|-------|-------|
| Stud Width   |      | Flange Height |      | Uncoated Steel Thk. | Stud Spacing | 5 psf (240Pa) Uniform Load |                  |       |       |
| inch         | (mm) | inch          | (mm) | mm                  | inch         | (mm)                       | Deflection Limit |       |       |
|              |      |               |      |                     |              |                            | L/120            | L/240 | L/360 |
| 2            | 50   | 1¼            | 32   | 0.8                 | 12           | 305                        | 4180             | 3320  | 2900  |
| 2            | 50   | 1¼            | 32   | 0.8                 | 16           | 406                        | 3800             | 3020  | 2640  |
| 2            | 50   | 1¼            | 32   | 0.8                 | 24           | 610                        | 3100             | 2630  | 2300  |
| 2½           | 64   | 1¼            | 32   | 0.8                 | 12           | 305                        | 5020             | 3980  | 3480  |
| 2½           | 64   | 1¼            | 32   | 0.8                 | 16           | 406                        | 4360             | 3620  | 3160  |
| 2½           | 64   | 1¼            | 32   | 0.8                 | 24           | 610                        | 3560             | 3160  | 2760  |
| 3            | 76   | 1¼            | 32   | 0.8                 | 12           | 305                        | 5580             | 4580  | 4000  |
| 3            | 76   | 1¼            | 32   | 0.8                 | 16           | 406                        | 4830             | 4160  | 3630  |
| 3            | 76   | 1¼            | 32   | 0.8                 | 24           | 610                        | 3940             | 3630  | 3170  |
| 3⅝           | 92   | 1¼            | 32   | 0.8                 | 12           | 305                        | 6200             | 5300  | 4630  |
| 3⅝           | 92   | 1¼            | 32   | 0.8                 | 16           | 406                        | 5380             | 4820  | 4210  |
| 3⅝           | 92   | 1¼            | 32   | 0.8                 | 24           | 610                        | 4950             | 4250  | 3710  |
| 4            | 102  | 1¼            | 32   | 0.8                 | 12           | 305                        | 6550             | 5720  | 5000  |
| 4            | 102  | 1¼            | 32   | 0.8                 | 16           | 406                        | 5680             | 5200  | 4540  |
| 4            | 102  | 1¼            | 32   | 0.8                 | 24           | 610                        | 4630             | 4540  | 3970  |



| 0.8mm member |      |               |      |                     | Limiting heights (mm)      |       |       |      |      |
|--------------|------|---------------|------|---------------------|----------------------------|-------|-------|------|------|
| Stud Width   |      | Flange Height |      | Uncoated Steel Thk. | 5 psf (240Pa) Uniform Load |       |       |      |      |
| inch         | (mm) | inch          | (mm) | mm                  | Deflection Limit           |       |       |      |      |
|              |      |               |      |                     | L/120                      | L/240 | L/360 |      |      |
| 2            | 50   | 1¾            | 45   | 0.8                 | 12                         | 305   | 4510  | 3580 | 3130 |
| 2            | 50   | 1¾            | 45   | 0.8                 | 16                         | 406   | 3970  | 3250 | 2840 |
| 2            | 50   | 1¾            | 45   | 0.8                 | 24                         | 610   | 3240  | 2840 | 2480 |
| 2½           | 64   | 1¾            | 45   | 0.8                 | 12                         | 305   | 5240  | 4260 | 3720 |
| 2½           | 64   | 1¾            | 45   | 0.8                 | 16                         | 406   | 4540  | 3880 | 3390 |
| 2½           | 64   | 1¾            | 45   | 0.8                 | 24                         | 610   | 3700  | 3380 | 2960 |
| 3            | 76   | 1¾            | 45   | 0.8                 | 12                         | 305   | 5770  | 4890 | 4270 |
| 3            | 76   | 1¾            | 45   | 0.8                 | 16                         | 406   | 5000  | 4450 | 3890 |
| 3            | 76   | 1¾            | 45   | 0.8                 | 24                         | 610   | 4080  | 3880 | 3390 |
| 3⅝           | 92   | 1¾            | 45   | 0.8                 | 12                         | 305   | 6390  | 5660 | 4950 |
| 3⅝           | 92   | 1¾            | 45   | 0.8                 | 16                         | 406   | 5540  | 5150 | 4500 |
| 3⅝           | 92   | 1¾            | 45   | 0.8                 | 24                         | 610   | 4520  | 4500 | 3930 |
| 4            | 102  | 1¾            | 45   | 0.8                 | 12                         | 305   | 6730  | 6110 | 5340 |
| 4            | 102  | 1¾            | 45   | 0.8                 | 16                         | 406   | 5840  | 5560 | 4850 |
| 4            | 102  | 1¾            | 45   | 0.8                 | 24                         | 610   | 4760  | 4760 | 4240 |
| 5            | 127  | 1¾            | 45   | 0.8                 | 12                         | 305   | 7590  | 7280 | 6360 |
| 5            | 127  | 1¾            | 45   | 0.8                 | 16                         | 406   | 6580  | 6580 | 5780 |
| 5            | 127  | 1¾            | 45   | 0.8                 | 24                         | 610   | 5370  | 5370 | 5050 |
| 6            | 152  | 1¾            | 45   | 0.8                 | 12                         | 305   | 8350  | 8350 | 7340 |
| 6            | 152  | 1¾            | 45   | 0.8                 | 16                         | 406   | 7240  | 7240 | 6670 |
| 6            | 152  | 1¾            | 45   | 0.8                 | 24                         | 610   | 5900  | 5900 | 5820 |

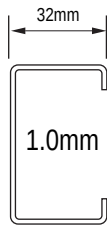


| 0.8mm member |      |               |      |                     |              | Limiting heights (mm) |                            |       |       |
|--------------|------|---------------|------|---------------------|--------------|-----------------------|----------------------------|-------|-------|
| Stud Width   |      | Flange Height |      | Uncoated Steel Thk. | Stud Spacing |                       | 5 psf (240Pa) Uniform Load |       |       |
| inch         | (mm) | inch          | (mm) | mm                  | inch         | (mm)                  | Deflection Limit           |       |       |
|              |      |               |      |                     |              |                       | L/120                      | L/240 | L/360 |
| 3⅝           | 92   | 2             | 50   | 0.8                 | 12           | 305                   | 6430                       | 5730  | 5010  |
| 3⅝           | 92   | 2             | 50   | 0.8                 | 16           | 406                   | 5570                       | 5210  | 4550  |
| 3⅝           | 92   | 2             | 50   | 0.8                 | 24           | 610                   | 4540                       | 4540  | 3970  |
| 4            | 102  | 2             | 50   | 0.8                 | 12           | 305                   | 6770                       | 6190  | 5400  |
| 4            | 102  | 2             | 50   | 0.8                 | 16           | 406                   | 5870                       | 5620  | 4910  |
| 4            | 102  | 2             | 50   | 0.8                 | 24           | 610                   | 4790                       | 4790  | 4290  |
| 5            | 127  | 2             | 50   | 0.8                 | 12           | 305                   | 7630                       | 7370  | 6440  |
| 5            | 127  | 2             | 50   | 0.8                 | 16           | 406                   | 6610                       | 6610  | 5850  |
| 5            | 127  | 2             | 50   | 0.8                 | 24           | 610                   | 5390                       | 5390  | 5110  |
| 6            | 152  | 2             | 50   | 0.8                 | 12           | 305                   | 8400                       | 8400  | 7450  |
| 6            | 152  | 2             | 50   | 0.8                 | 16           | 406                   | 7280                       | 7280  | 6770  |
| 6            | 152  | 2             | 50   | 0.8                 | 24           | 610                   | 5940                       | 5940  | 5910  |

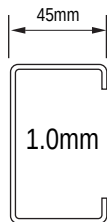


## Limiting Heights:

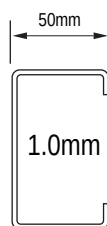
1. Limiting heights computed are based on the theoretical properties of steel stud and deflection limits of L/120; L/240; and L/360 resulted from a uniform load of 5psf (240Pa) on partition surface, per ASTM C754 and GA-600 Gypsum Association.
2. Limiting heights are in accordance with AISI Standard "North American Specification For The Design of Cold-formed Steel" using all steel non-composite design.
3. Limiting heights are established by considering flexure, shear, web crippling and deflection, based on continuous support of each flange over the full length of the stud; with unperforated section properties; and full lateral support to both flanges in flexure.
4. Applications involving multiple spans, cantilevers, concentrated loads, etc should be investigated separately.



| 1.0mm member |      |               |      |                     | Limiting heights (mm)      |       |       |
|--------------|------|---------------|------|---------------------|----------------------------|-------|-------|
| Stud Width   |      | Flange Height |      | Uncoated Steel Thk. | 5 psf (240Pa) Uniform Load |       |       |
| inch         | (mm) | inch          | (mm) | mm                  | Deflection Limit           |       |       |
|              |      |               |      |                     | L/120                      | L/240 | L/360 |
| 2            | 50   | 1 1/4         | 32   | 1.0                 | 4480                       | 3560  | 3110  |
| 2            | 50   | 1 1/4         | 32   | 1.0                 | 4080                       | 3230  | 2830  |
| 2            | 50   | 1 1/4         | 32   | 1.0                 | 3530                       | 2820  | 2470  |
| 2 1/2        | 64   | 1 1/4         | 32   | 1.0                 | 5380                       | 4270  | 3730  |
| 2 1/2        | 64   | 1 1/4         | 32   | 1.0                 | 4890                       | 3880  | 3390  |
| 2 1/2        | 64   | 1 1/4         | 32   | 1.0                 | 4100                       | 3390  | 2960  |
| 3            | 76   | 1 1/4         | 32   | 1.0                 | 6180                       | 4910  | 4290  |
| 3            | 76   | 1 1/4         | 32   | 1.0                 | 5590                       | 4460  | 3900  |
| 3            | 76   | 1 1/4         | 32   | 1.0                 | 4560                       | 3900  | 3400  |
| 3 5/8        | 92   | 1 1/4         | 32   | 1.0                 | 7160                       | 5690  | 4970  |
| 3 5/8        | 92   | 1 1/4         | 32   | 1.0                 | 6240                       | 5170  | 4520  |
| 3 5/8        | 92   | 1 1/4         | 32   | 1.0                 | 5090                       | 4510  | 3940  |
| 4            | 102  | 1 1/4         | 32   | 1.0                 | 7620                       | 6140  | 5360  |
| 4            | 102  | 1 1/4         | 32   | 1.0                 | 6600                       | 5580  | 4880  |
| 4            | 102  | 1 1/4         | 32   | 1.0                 | 5390                       | 4870  | 4260  |



| 1.0mm member |      |               |      |                     | Limiting heights (mm)      |       |       |
|--------------|------|---------------|------|---------------------|----------------------------|-------|-------|
| Stud Width   |      | Flange Height |      | Uncoated Steel Thk. | 5 psf (240Pa) Uniform Load |       |       |
| inch         | (mm) | inch          | (mm) | mm                  | Deflection Limit           |       |       |
|              |      |               |      |                     | L/120                      | L/240 | L/360 |
| 2            | 50   | 1 3/4         | 45   | 1.0                 | 4880                       | 3870  | 3380  |
| 2            | 50   | 1 3/4         | 45   | 1.0                 | 4430                       | 3520  | 3070  |
| 2            | 50   | 1 3/4         | 45   | 1.0                 | 3730                       | 3070  | 2680  |
| 2 1/2        | 64   | 1 3/4         | 45   | 1.0                 | 5820                       | 4620  | 4040  |
| 2 1/2        | 64   | 1 3/4         | 45   | 1.0                 | 5270                       | 4200  | 3670  |
| 2 1/2        | 64   | 1 3/4         | 45   | 1.0                 | 4300                       | 3670  | 3200  |
| 3            | 76   | 1 3/4         | 45   | 1.0                 | 6670                       | 5290  | 4620  |
| 3            | 76   | 1 3/4         | 45   | 1.0                 | 5830                       | 4810  | 4200  |
| 3            | 76   | 1 3/4         | 45   | 1.0                 | 4750                       | 4200  | 3670  |
| 3 5/8        | 92   | 1 3/4         | 45   | 1.0                 | 7460                       | 6110  | 5340  |
| 3 5/8        | 92   | 1 3/4         | 45   | 1.0                 | 6470                       | 5550  | 4850  |
| 3 5/8        | 92   | 1 3/4         | 45   | 1.0                 | 5280                       | 4850  | 4240  |
| 4            | 102  | 1 3/4         | 45   | 1.0                 | 7880                       | 6580  | 5750  |
| 4            | 102  | 1 3/4         | 45   | 1.0                 | 6830                       | 5990  | 5230  |
| 4            | 102  | 1 3/4         | 45   | 1.0                 | 5570                       | 5230  | 4570  |
| 5            | 127  | 1 3/4         | 45   | 1.0                 | 8910                       | 7820  | 6840  |
| 5            | 127  | 1 3/4         | 45   | 1.0                 | 7720                       | 7110  | 6210  |
| 5            | 127  | 1 3/4         | 45   | 1.0                 | 6300                       | 6210  | 5430  |
| 6            | 152  | 1 3/4         | 45   | 1.0                 | 9830                       | 9030  | 7890  |
| 6            | 152  | 1 3/4         | 45   | 1.0                 | 8520                       | 8210  | 7170  |
| 6            | 152  | 1 3/4         | 45   | 1.0                 | 6950                       | 6950  | 6260  |



| 1.0mm member |      |               |      |                     | Limiting heights (mm)      |       |       |
|--------------|------|---------------|------|---------------------|----------------------------|-------|-------|
| Stud Width   |      | Flange Height |      | Uncoated Steel Thk. | 5 psf (240Pa) Uniform Load |       |       |
| inch         | (mm) | inch          | (mm) | mm                  | Deflection Limit           |       |       |
|              |      |               |      |                     | L/120                      | L/240 | L/360 |
| 3 5/8        | 92   | 2             | 50   | 1.0                 | 7520                       | 6260  | 5470  |
| 3 5/8        | 92   | 2             | 50   | 1.0                 | 6520                       | 5690  | 4970  |
| 3 5/8        | 92   | 2             | 50   | 1.0                 | 5320                       | 4970  | 4340  |
| 4            | 102  | 2             | 50   | 1.0                 | 7930                       | 6740  | 5890  |
| 4            | 102  | 2             | 50   | 1.0                 | 6880                       | 6130  | 5360  |
| 4            | 102  | 2             | 50   | 1.0                 | 5610                       | 5350  | 4680  |
| 5            | 127  | 2             | 50   | 1.0                 | 8960                       | 8000  | 6990  |
| 5            | 127  | 2             | 50   | 1.0                 | 7770                       | 7270  | 6350  |
| 5            | 127  | 2             | 50   | 1.0                 | 6340                       | 6340  | 5550  |
| 6            | 152  | 2             | 50   | 1.0                 | 9900                       | 9220  | 8050  |
| 6            | 152  | 2             | 50   | 1.0                 | 8580                       | 8380  | 7320  |
| 6            | 152  | 2             | 50   | 1.0                 | 7000                       | 7000  | 6390  |

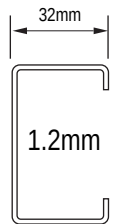




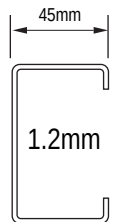
## Limiting Heights:

1. Limiting heights computed are based on the theoretical properties of steel stud and deflection limits of L/120; L/240; and L/360 resulted from a uniform load of 5psf (240Pa) on partition surface, per ASTM C754 and GA-600 Gypsum Association.
2. Limiting heights are in accordance with AISI Standard "North American Specification For The Design of Cold-formed Steel" using all steel non-composite design.
3. Limiting heights are established by considering flexure, shear, web crippling and deflection, based on continuous support of each flange over the full length of the stud; with unperforated section properties; and full lateral support to both flanges in flexure.
4. Applications involving multiple spans, cantilevers, concentrated loads, etc should be investigated separately.

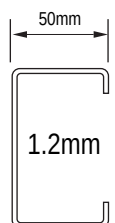
| 1.2mm member |      |               |      |                     |              | Limiting heights (mm)      |                  |       |       |
|--------------|------|---------------|------|---------------------|--------------|----------------------------|------------------|-------|-------|
| Stud Width   |      | Flange Height |      | Uncoated Steel Thk. | Stud Spacing | 5 psf (240Pa) Uniform Load |                  |       |       |
| inch         | (mm) | inch          | (mm) | mm                  | inch         | (mm)                       | Deflection Limit |       |       |
|              |      |               |      |                     |              |                            | L/120            | L/240 | L/360 |
| 2            | 50   | 1¼            | 32   | 1.2                 | 12           | 305                        | 4740             | 3760  | 3280  |
| 2            | 50   | 1¼            | 32   | 1.2                 | 16           | 406                        | 4310             | 3420  | 2990  |
| 2            | 50   | 1¼            | 32   | 1.2                 | 24           | 610                        | 3760             | 2980  | 2610  |
| 2½           | 64   | 1¼            | 32   | 1.2                 | 12           | 305                        | 5680             | 4510  | 3940  |
| 2½           | 64   | 1¼            | 32   | 1.2                 | 16           | 406                        | 5170             | 4100  | 3580  |
| 2½           | 64   | 1¼            | 32   | 1.2                 | 24           | 610                        | 4470             | 3580  | 3130  |
| 3            | 76   | 1¼            | 32   | 1.2                 | 12           | 305                        | 6540             | 5190  | 4540  |
| 3            | 76   | 1¼            | 32   | 1.2                 | 16           | 406                        | 5950             | 4720  | 4120  |
| 3            | 76   | 1¼            | 32   | 1.2                 | 24           | 610                        | 5040             | 4120  | 3600  |
| 3⅝           | 92   | 1¼            | 32   | 1.2                 | 12           | 305                        | 7580             | 6020  | 5260  |
| 3⅝           | 92   | 1¼            | 32   | 1.2                 | 16           | 406                        | 6890             | 5470  | 4780  |
| 3⅝           | 92   | 1¼            | 32   | 1.2                 | 24           | 610                        | 5720             | 4780  | 4170  |
| 4            | 102  | 1¼            | 32   | 1.2                 | 12           | 305                        | 8190             | 6500  | 5680  |
| 4            | 102  | 1¼            | 32   | 1.2                 | 16           | 406                        | 7440             | 5910  | 5160  |
| 4            | 102  | 1¼            | 32   | 1.2                 | 24           | 610                        | 6070             | 5160  | 4510  |



| 1.2mm member |      |               |      |                     |              | Limiting heights (mm)      |                  |       |       |
|--------------|------|---------------|------|---------------------|--------------|----------------------------|------------------|-------|-------|
| Stud Width   |      | Flange Height |      | Uncoated Steel Thk. | Stud Spacing | 5 psf (240Pa) Uniform Load |                  |       |       |
| inch         | (mm) | inch          | (mm) | mm                  | inch         | (mm)                       | Deflection Limit |       |       |
|              |      |               |      |                     |              |                            | L/120            | L/240 | L/360 |
| 2            | 50   | 1¾            | 45   | 1.2                 | 12           | 305                        | 5160             | 4100  | 3580  |
| 2            | 50   | 1¾            | 45   | 1.2                 | 16           | 406                        | 4690             | 3730  | 3250  |
| 2            | 50   | 1¾            | 45   | 1.2                 | 24           | 610                        | 4100             | 3250  | 2840  |
| 2½           | 64   | 1¾            | 45   | 1.2                 | 12           | 305                        | 6170             | 4890  | 4270  |
| 2½           | 64   | 1¾            | 45   | 1.2                 | 16           | 406                        | 5600             | 4450  | 3890  |
| 2½           | 64   | 1¾            | 45   | 1.2                 | 24           | 610                        | 4850             | 3880  | 3390  |
| 3            | 76   | 1¾            | 45   | 1.2                 | 12           | 305                        | 7070             | 5610  | 4900  |
| 3            | 76   | 1¾            | 45   | 1.2                 | 16           | 406                        | 6420             | 5100  | 4450  |
| 3            | 76   | 1¾            | 45   | 1.2                 | 24           | 610                        | 5370             | 4450  | 3890  |
| 3⅝           | 92   | 1¾            | 45   | 1.2                 | 12           | 305                        | 8160             | 6480  | 5660  |
| 3⅝           | 92   | 1¾            | 45   | 1.2                 | 16           | 406                        | 7330             | 5890  | 5140  |
| 3⅝           | 92   | 1¾            | 45   | 1.2                 | 24           | 610                        | 5980             | 5140  | 4490  |
| 4            | 102  | 1¾            | 45   | 1.2                 | 12           | 305                        | 8800             | 6980  | 6100  |
| 4            | 102  | 1¾            | 45   | 1.2                 | 16           | 406                        | 7750             | 6350  | 5540  |
| 4            | 102  | 1¾            | 45   | 1.2                 | 24           | 610                        | 6320             | 5540  | 4840  |
| 5            | 127  | 1¾            | 45   | 1.2                 | 12           | 305                        | 10140            | 8300  | 7250  |
| 5            | 127  | 1¾            | 45   | 1.2                 | 16           | 406                        | 8790             | 7550  | 6590  |
| 5            | 127  | 1¾            | 45   | 1.2                 | 24           | 610                        | 7170             | 6590  | 5760  |
| 6            | 152  | 1¾            | 45   | 1.2                 | 12           | 305                        | 11000            | 9590  | 8370  |
| 6            | 152  | 1¾            | 45   | 1.2                 | 16           | 406                        | 9730             | 8710  | 7610  |
| 6            | 152  | 1¾            | 45   | 1.2                 | 24           | 610                        | 7940             | 7610  | 6650  |



| 1.2mm member |      |               |      |                     |              | Limiting heights (mm) |                            |       |       |
|--------------|------|---------------|------|---------------------|--------------|-----------------------|----------------------------|-------|-------|
| Stud Width   |      | Flange Height |      | Uncoated Steel Thk. | Stud Spacing |                       | 5 psf (240Pa) Uniform Load |       |       |
| inch         | (mm) | inch          | (mm) | mm                  | inch         | (mm)                  | Deflection Limit           |       |       |
|              |      |               |      |                     |              |                       | L/120                      | L/240 | L/360 |
| 3⅝           | 92   | 2             | 50   | 1.2                 | 12           | 305                   | 8360                       | 6640  | 5800  |
| 3⅝           | 92   | 2             | 50   | 1.2                 | 16           | 406                   | 7400                       | 6030  | 5270  |
| 3⅝           | 92   | 2             | 50   | 1.2                 | 24           | 610                   | 6040                       | 5270  | 4600  |
| 4            | 102  | 2             | 50   | 1.2                 | 12           | 305                   | 9010                       | 7150  | 6240  |
| 4            | 102  | 2             | 50   | 1.2                 | 16           | 406                   | 7820                       | 6500  | 5680  |
| 4            | 102  | 2             | 50   | 1.2                 | 24           | 610                   | 6380                       | 5670  | 4960  |
| 5            | 127  | 2             | 50   | 1.2                 | 12           | 305                   | 10220                      | 8480  | 7410  |
| 5            | 127  | 2             | 50   | 1.2                 | 16           | 406                   | 8860                       | 7710  | 6740  |
| 5            | 127  | 2             | 50   | 1.2                 | 24           | 610                   | 7230                       | 6730  | 5880  |
| 6            | 152  | 2             | 50   | 1.2                 | 12           | 305                   | 11000                      | 9780  | 8540  |
| 6            | 152  | 2             | 50   | 1.2                 | 16           | 406                   | 9800                       | 8890  | 7770  |
| 6            | 152  | 2             | 50   | 1.2                 | 24           | 610                   | 8000                       | 7760  | 6780  |

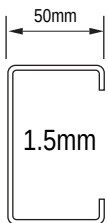


## Limiting Heights:

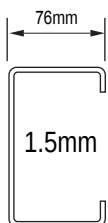
1. Limiting heights computed are based on the theoretical properties of steel stud and deflection limits of L/120; L/240; and L/360 resulted from a uniform load of 5psf (240Pa) on partition surface, per ASTM C754 and GA-600 Gypsum Association.
2. Limiting heights are in accordance with AISI Standard "North American Specification For The Design of Cold-formed Steel" using all steel non-composite design.
3. Limiting heights are established by considering flexure, shear, web crippling and deflection, based on continuous support of each flange over the full length of the stud; with unperforated section properties; and full lateral support to both flanges in flexure.
4. Applications involving multiple spans, cantilevers, concentrated loads, etc should be investigated separately.



| 1.5mm member |      |               |      |                     | Limiting heights (mm)      |       |       |
|--------------|------|---------------|------|---------------------|----------------------------|-------|-------|
| Stud Width   |      | Flange Height |      | Uncoated Steel Thk. | 5 psf (240Pa) Uniform Load |       |       |
| inch         | (mm) | inch          | (mm) | mm                  | Deflection Limit           |       |       |
|              |      |               |      |                     | L/120                      | L/240 | L/360 |
| 2            | 50   | 1 3/4         | 45   | 1.5                 | 5540                       | 4390  | 3840  |
| 2            | 50   | 1 3/4         | 45   | 1.5                 | 5030                       | 4000  | 3490  |
| 2            | 50   | 1 3/4         | 45   | 1.5                 | 4390                       | 3490  | 3050  |
| 2 1/2        | 64   | 1 3/4         | 45   | 1.5                 | 6620                       | 5250  | 4590  |
| 2 1/2        | 64   | 1 3/4         | 45   | 1.5                 | 6020                       | 4780  | 4170  |
| 2 1/2        | 64   | 1 3/4         | 45   | 1.5                 | 5250                       | 4170  | 3640  |
| 3            | 76   | 1 3/4         | 45   | 1.5                 | 7590                       | 6030  | 5260  |
| 3            | 76   | 1 3/4         | 45   | 1.5                 | 6900                       | 5480  | 4790  |
| 3            | 76   | 1 3/4         | 45   | 1.5                 | 6030                       | 4780  | 4180  |
| 3 5/8        | 92   | 1 3/4         | 45   | 1.5                 | 8770                       | 6960  | 6080  |
| 3 5/8        | 92   | 1 3/4         | 45   | 1.5                 | 7970                       | 6330  | 5530  |
| 3 5/8        | 92   | 1 3/4         | 45   | 1.5                 | 6940                       | 5530  | 4830  |
| 4            | 102  | 1 3/4         | 45   | 1.5                 | 9460                       | 7510  | 6560  |
| 4            | 102  | 1 3/4         | 45   | 1.5                 | 8600                       | 6830  | 5960  |
| 4            | 102  | 1 3/4         | 45   | 1.5                 | 7350                       | 5960  | 5210  |
| 5            | 127  | 1 3/4         | 45   | 1.5                 | 10900                      | 8940  | 7810  |
| 5            | 127  | 1 3/4         | 45   | 1.5                 | 10240                      | 8130  | 7100  |
| 5            | 127  | 1 3/4         | 45   | 1.5                 | 8380                       | 7090  | 6200  |
| 6            | 152  | 1 3/4         | 45   | 1.5                 | 11000                      | 10320 | 9010  |
| 6            | 152  | 1 3/4         | 45   | 1.5                 | 11000                      | 9380  | 8190  |
| 6            | 152  | 1 3/4         | 45   | 1.5                 | 9320                       | 8190  | 7150  |



| 1.5mm member |      |               |      |                     | Limiting heights (mm)      |       |       |
|--------------|------|---------------|------|---------------------|----------------------------|-------|-------|
| Stud Width   |      | Flange Height |      | Uncoated Steel Thk. | 5 psf (240Pa) Uniform Load |       |       |
| inch         | (mm) | inch          | (mm) | mm                  | Deflection Limit           |       |       |
|              |      |               |      |                     | L/120                      | L/240 | L/360 |
| 3 5/8        | 92   | 2             | 50   | 1.5                 | 8970                       | 7120  | 6220  |
| 3 5/8        | 92   | 2             | 50   | 1.5                 | 8160                       | 6470  | 5650  |
| 3 5/8        | 92   | 2             | 50   | 1.5                 | 7030                       | 5650  | 4940  |
| 4            | 102  | 2             | 50   | 1.5                 | 9670                       | 7670  | 6700  |
| 4            | 102  | 2             | 50   | 1.5                 | 8790                       | 6980  | 6090  |
| 4            | 102  | 2             | 50   | 1.5                 | 7440                       | 6090  | 5320  |
| 5            | 127  | 2             | 50   | 1.5                 | 11000                      | 9110  | 7960  |
| 5            | 127  | 2             | 50   | 1.5                 | 10380                      | 8280  | 7240  |
| 5            | 127  | 2             | 50   | 1.5                 | 8470                       | 7230  | 6320  |
| 6            | 152  | 2             | 50   | 1.5                 | 11000                      | 10510 | 9180  |
| 6            | 152  | 2             | 50   | 1.5                 | 11000                      | 9550  | 8350  |
| 6            | 152  | 2             | 50   | 1.5                 | 9410                       | 8340  | 7290  |



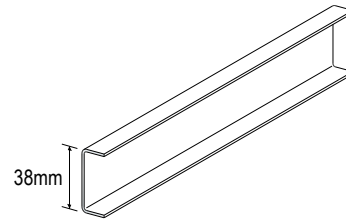
| 1.5mm member |      |               |      |                     | Limiting heights (mm)      |       |       |
|--------------|------|---------------|------|---------------------|----------------------------|-------|-------|
| Stud Width   |      | Flange Height |      | Uncoated Steel Thk. | 5 psf (240Pa) Uniform Load |       |       |
| inch         | (mm) | inch          | (mm) | mm                  | Deflection Limit           |       |       |
|              |      |               |      |                     | L/120                      | L/240 | L/360 |
| 4            | 102  | 3             | 76   | 1.5                 | N/A                        | 8750  | 7650  |
| 4            | 102  | 3             | 76   | 1.5                 | N/A                        | 7960  | 6950  |
| 4            | 102  | 3             | 76   | 1.5                 | N/A                        | 6950  | 6070  |
| 5            | 127  | 3             | 76   | 1.5                 | N/A                        | 10340 | 9030  |
| 5            | 127  | 3             | 76   | 1.5                 | N/A                        | 9400  | 8210  |
| 5            | 127  | 3             | 76   | 1.5                 | N/A                        | 8200  | 7170  |
| 6            | 152  | 3             | 76   | 1.5                 | N/A                        | 11000 | 10370 |
| 6            | 152  | 3             | 76   | 1.5                 | N/A                        | 10790 | 9420  |
| 6            | 152  | 3             | 76   | 1.5                 | N/A                        | 9420  | 8230  |

## Lateral Bracing

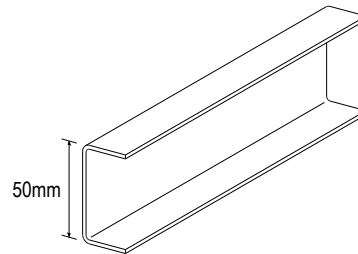
The purpose of bracing is to provide resistance to lateral and torsional movement of the stud assembly. Bracing is normally required when:

- Walls are only sheathed on one side.
- Wall sheathing does not extend to the underside of the concrete slab in high wall assembly.
- To brace studs when using single deflection channel system.
- To increase the rigidity of the wall assembly.

**STUDCO** cold-rolled bracer channels are available in 38mm and 50mm widths, are manufactured from corrosion resistant galvanized steel. To brace steel studs laterally, the bracer channel is inserted through the punchouts of the framing members at specified vertical spacing. The 1½" x 1½" metal angle is then fastened to the framing member and the bracer channel with pan head screws per connection.

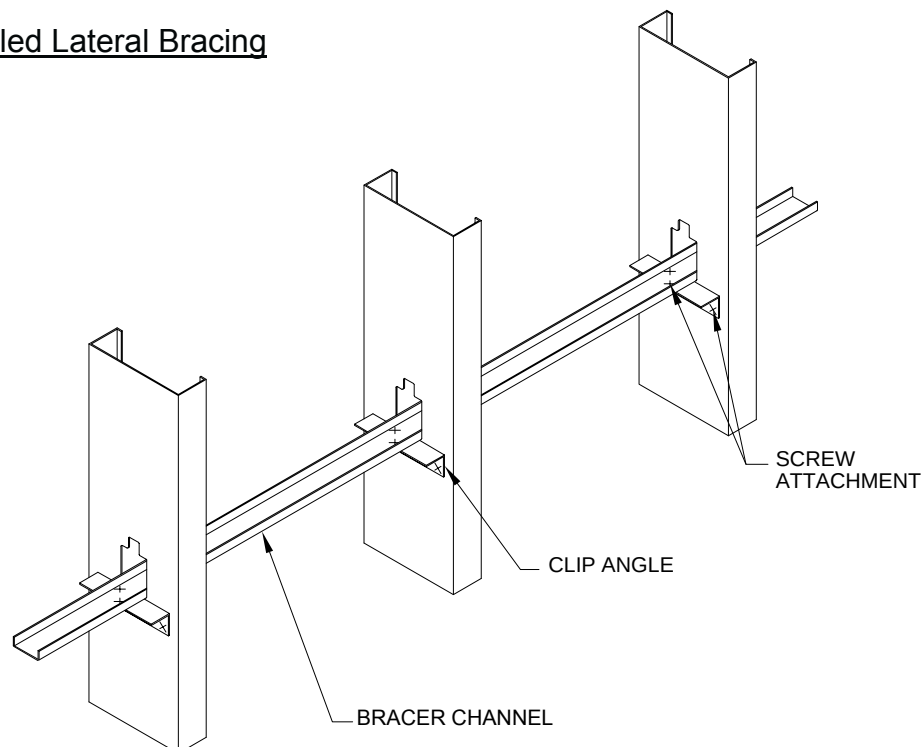


C-38 bracer channel is recommended for stud widths 2½", 3", 3⅝", 4", 5" and 6".



C-50 bracer channel is recommended for stud widths 6" and 8"

## Cold Rolled Lateral Bracing





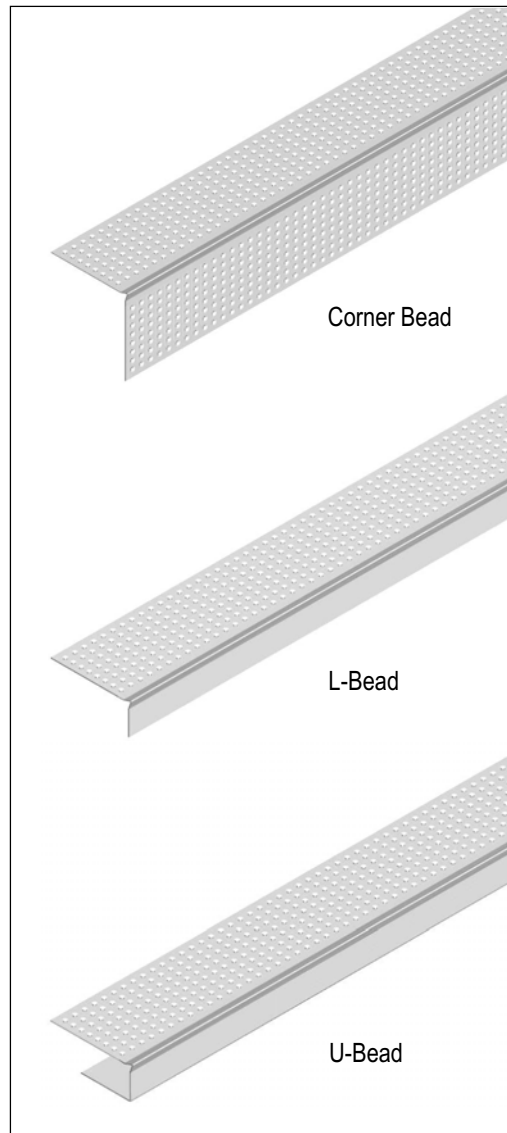
## Metal Beads

KIRII metal beads including Corner Bead, L-Bead and U-Bead are hot-dipped, highly durable galvanized beads that provide enhanced corrosion protection. Flanges are knurled and punched to ease joint compound adhesion and screw attachment. Products comply with the coating requirement of ASTM C1047 standard.

CORNER BEAD ( $1\frac{1}{4}$ " ) is used to provide a high-quality finish and solid protection at 90° drywall corners.

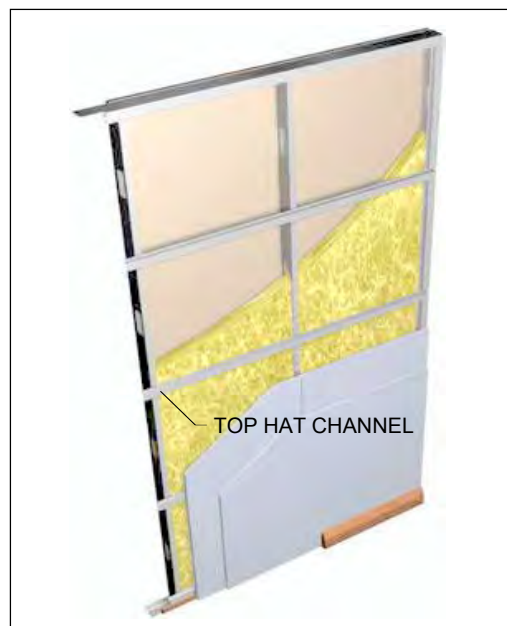
L-BEAD is used to provide a smooth finished edge for gypsum wallboard and it can be installed after board installation. Available in both  $\frac{1}{2}$ " and  $\frac{5}{8}$ " sizes.

U-BEAD is used to provide solid protection and smooth finished edges on gypsum wallboards around door and window openings. Available in both  $\frac{1}{2}$ " and  $\frac{5}{8}$ " sizes.



## Top Hat Channel

TOP HAT CHANNEL is manufactured from galvanized steel, is used for screw attachment to framing members. It provides a low cost method of reducing sound transmission in steel stud interior partition by suspending the gypsum wallboards away from the stud surface. Alternatively, it can be used to provide a sub-framing for gypsum board lining on masonry wall.





## Architectural Specifications

### **PART 1 - MATERIALS**

#### **A) FRAMING**

- 1) Non-loadbearing wall framing shall be **STUDCO** Partition Steel Framing System complying with ASTM C645 standard and manufactured by KIRII (HONG KONG) LIMITED.
- 2) C-shaped studs with 45mm height of flanges, hot-dip galvanized and shall be in one continuous length. Web of studs shall be punchouts for placing C-38 cold-rolled bracer channels at 1200mm c/c or as specified by the architect.
- 3) Runners shall be formed in U-shaped configuration, having web width compatible with those of the studs of the same nominal size. Besides, runners shall be designed such that when the studs are placed in both the top and bottom runners, they shall be held by friction. Height of flanges shall be 45mm or 32mm min.
- 4) Studs and Runners shall be manufactured from steel having a minimum uncoated steel thickness of 1.5mm and having a protective zinc coating mass of Z275/G90 to ASTM A653.

#### **B) GYPSUM BOARD**

- 1) Gypsum wallboard shall be 'ProRoc' Type C with tapered edges complying with BS EN 520 and ASTM C1396 standard and manufactured by Thai Gypsum Products Pcl.

### **PART 2 - INSTALLATION**

#### **A) FRAMING**

- 1) Accurately locate floor and ceiling runners along centerline of wall ensuring vertical alignment. Secure runners with suitable fasteners at 24" (610mm) maximum centers and 2" (50mm) from runner ends. At partition corners, one runner shall extend to the end of the corner and the other runner shall butt to it, allowing clearance for the gypsum board thickness as necessary. Do not miter runners.

- 2) Studs to be cut  $\frac{1}{4}$ " (6mm) short of the assembly height to accommodate floor variations. Position studs into runners and twist to secure by friction fit. Space studs at 24" (610mm) maximum centers with open sides facing the same direction. Studs located adjacent to door and window frames, partition intersection, and corners shall be anchored to runner flanges by screws or by crimping at each stud and runner flange.

#### **B) GYPSUM BOARD (SINGER LAYER)**

- 1) Apply gypsum board with the length parallel to the studs. Center abutting edges on stud flanges. Attached the gypsum board to studs with Type S drywall screws, they shall penetrate framing not less than  $\frac{3}{8}$ " (10mm) and be spaced a maximum of 12" (305mm) o.c. on all studs when the framing is 24" (610mm) o.c. Screws shall be spaced not less than  $\frac{3}{8}$ " (10mm) from ends and edges of gypsum board.
- 2) Starting in the upper corner of the wall, work across and down. Ends and edges of boards shall be touching but not forced tightly together.
- 3) Holding gypsum board firmly against the framing, begin fastening in the center of the board and move outward toward ends and edges.
- 4) Set fastener heads slightly indented from the surface without breaking the face paper or damaging the gypsum core.
- 5) Where required, install all services and insulation into the wall cavity before fixing boards to the second sides. Cut the first board to appropriate width so as to ensure that the joints are staggered on alternate studs to joints on the first side.
- 6) Cut openings out of full boards or lay out joints to fall on the studs nearest the center of doors and windows. Where possible, position full boards to extend completely above and below openings and avoid piecing.

#### **C) JOINT AND FASTENER TREATMENT**

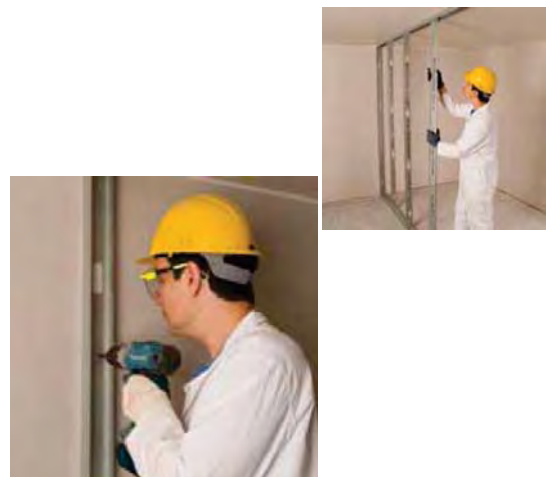
- 1) Joint compound and reinforcing tape shall be 'GypCote' finishing products complying with ASTM C475 standard.
- 2) Procedure shall be in accordance with manufacturer's recommendations.

## Construction details

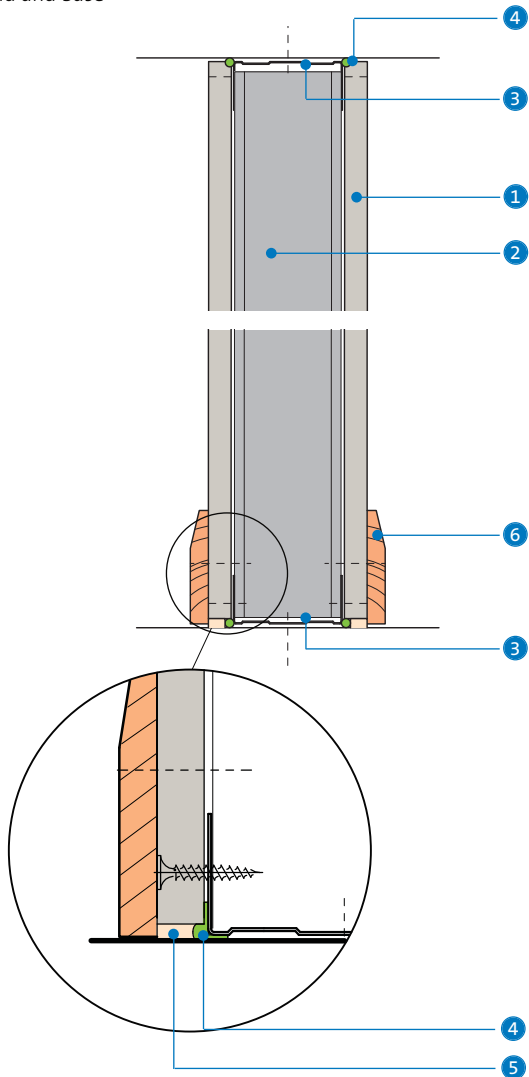
### 1 Steel stud framing



### 2 Steel stud framing



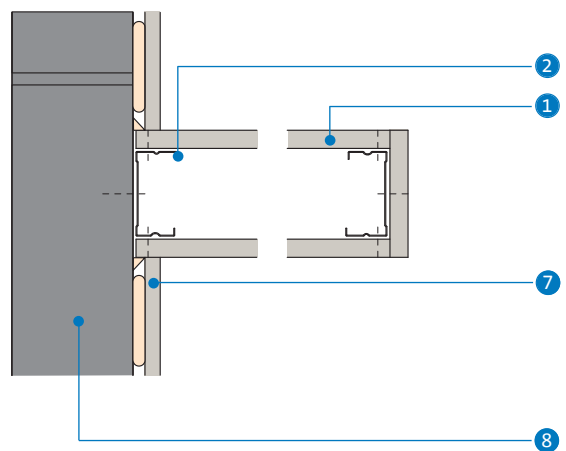
### 3 Head and base



### 4 Door opening



### 5 Junction with masonry and stop end detail

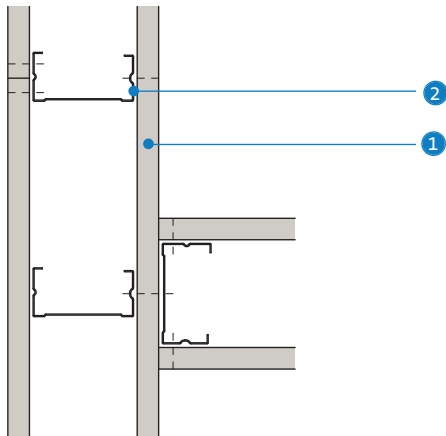


- ① Gyproc gypsum wallboard
- ② KIRII steel stud
- ③ KIRII top and bottom runner
- ④ Gyproc sealant or equivalent
- ⑤ Bulk fill with Gyproc jointing materials (where gap exceeds 5mm)
- ⑥ Skirting

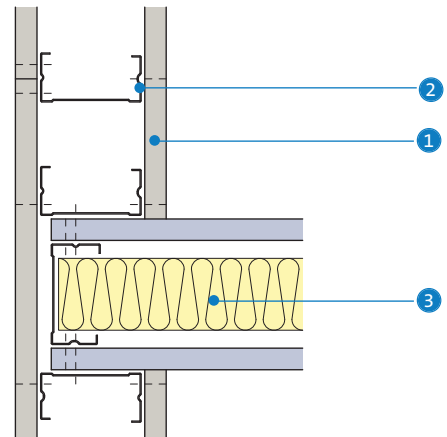
- ⑦ Wall lining system
- ⑧ Internal blockwork

## Construction details

6 'T' junction - single layer



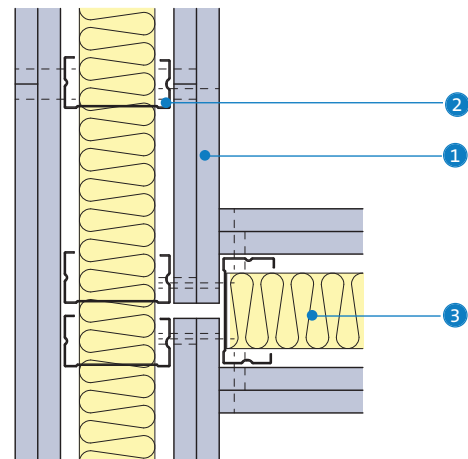
7 'T' Junction when partition with higher acoustic performance abuts a partition with lower acoustic performance. Acoustic principles only - detail may not be suitable for all solutions



8 Board fixing



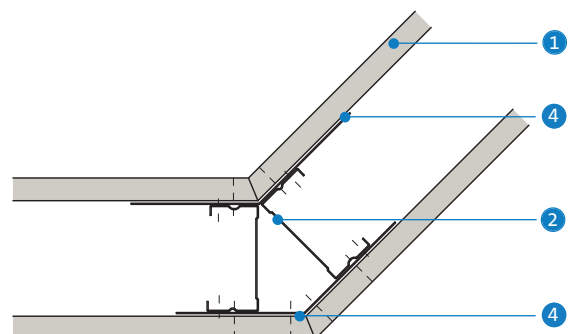
9 'T' junction to optimise acoustic performance and reduce flanking transmission



10 Board fixing



11 Splayed corner

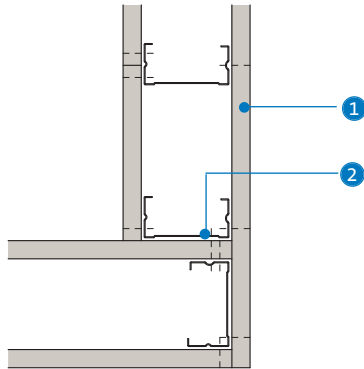


- 1 Gyproc gypsum wallboard
- 2 KIRII steel stud
- 3 Sound insulation
- 4 Splayed metal angle

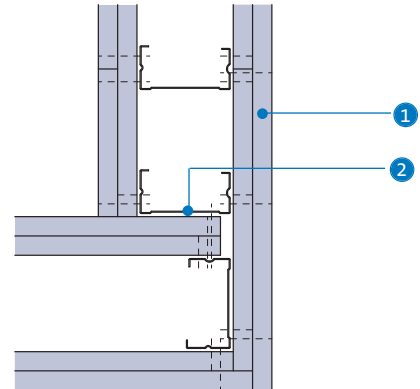


## Construction details

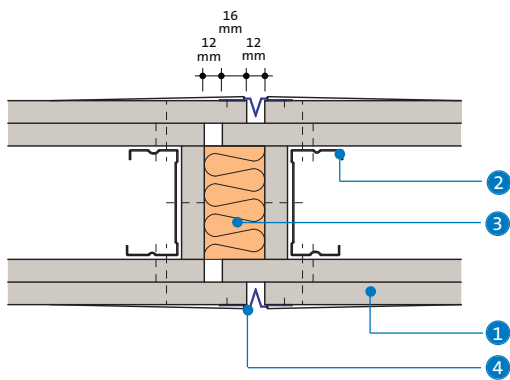
12 Corner detail - single layer



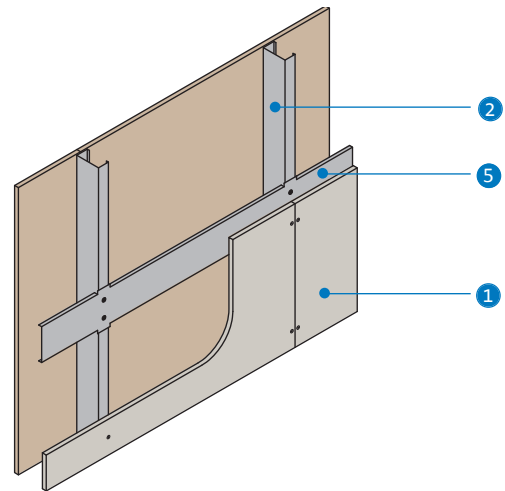
13 Corner detail - double layer



14 Typical control joint



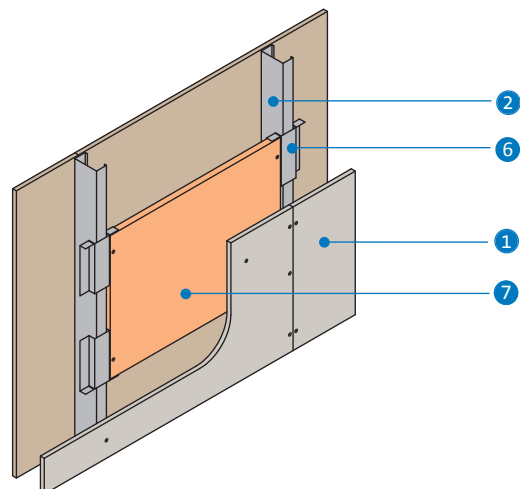
15 Fixing Strap for light weight fixtures



16 Fixtures & Services



17 Service Support Plate

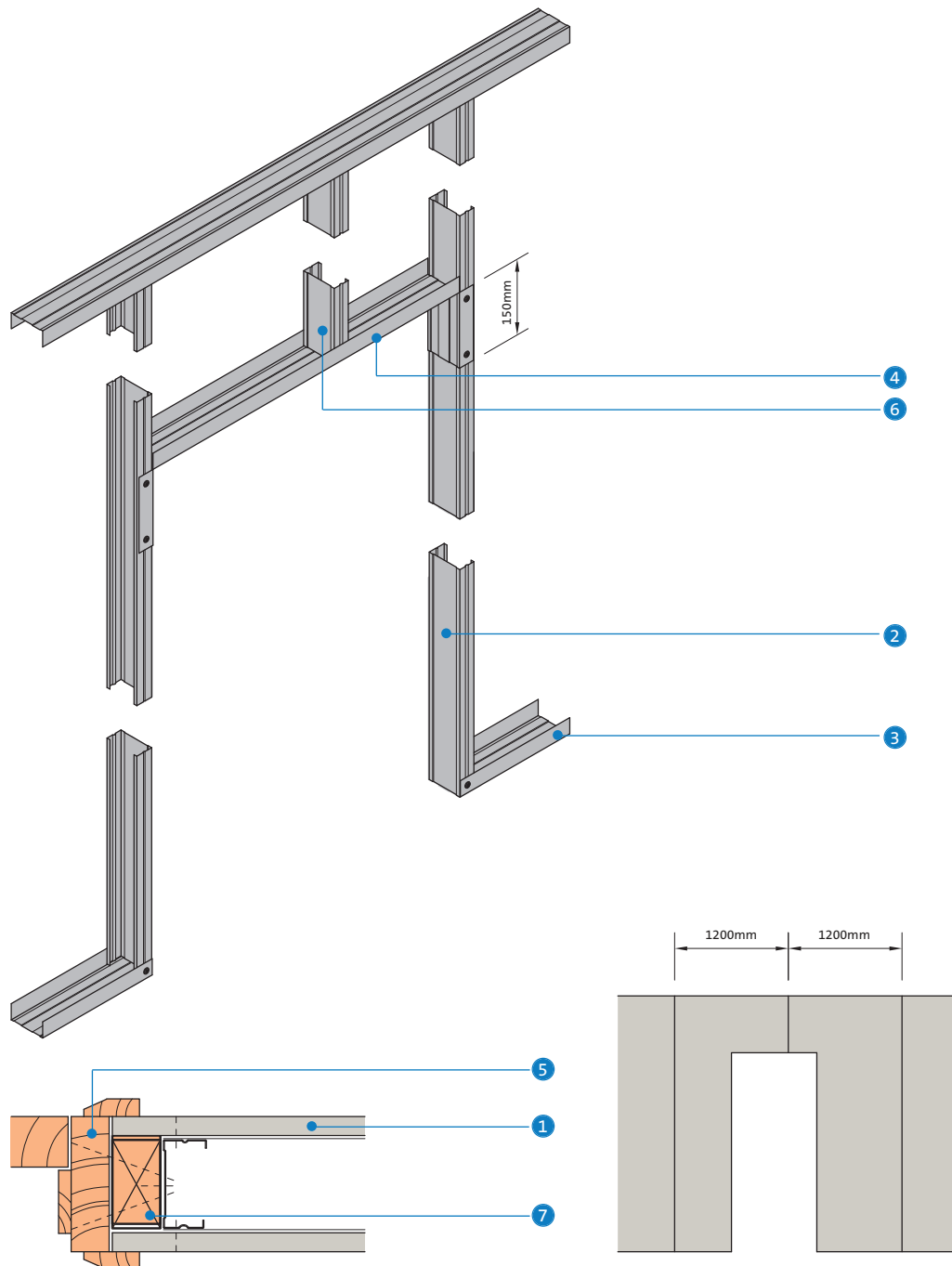


- ① Gyproc gypsum wallboard
- ② KIRII steel stud
- ③ Stone mineral wool
- ④ Gyproc control joint
- ⑤ Fixing strap (0.9mm)

- ⑥ Service support plate
- ⑦ 18mm plywood

## Construction details

### 18 Door frame



- |                                                              |                                            |
|--------------------------------------------------------------|--------------------------------------------|
| 1 Gyproc gypsum wallboard                                    | 5 Timber door frame and architrave         |
| 2 Kirii steel stud                                           | 6 Kirii steel stud to maintain stud module |
| 3 Kirii top and bottom runner                                | 7 Timber sub-frame                         |
| 4 Kirii top and bottom runner cut and bent to form door head |                                            |

**NB** Advice should be sought from the door manufacturer prior to the construction of these details.



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